Power

$$P = VI \checkmark$$

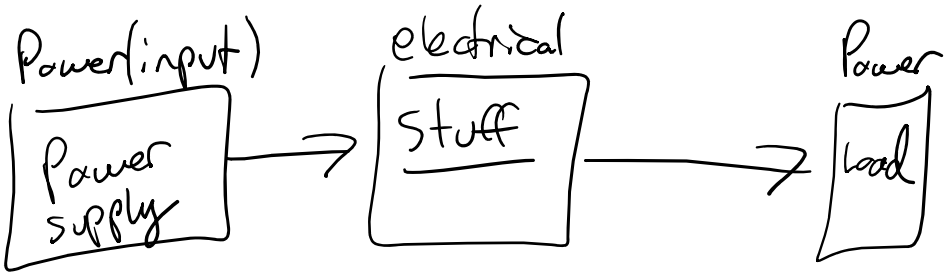
$$P = I^2 R$$

$$P = \frac{V^2}{R}$$

* Power is conserved

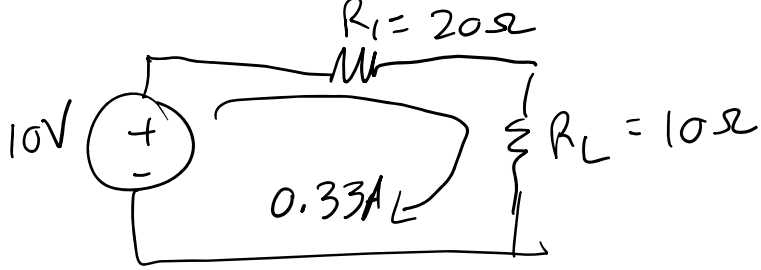
$$P_{IN} = P_{out}$$

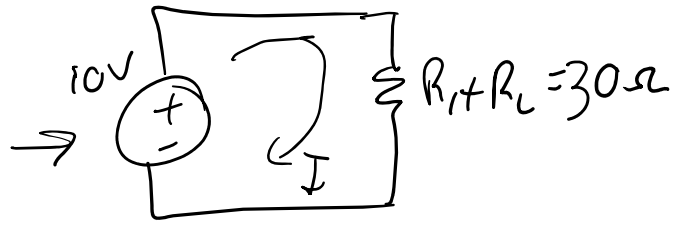
Efficiency is the ratio of Power that goes in to the power delivered (consumed) by the Load



$$\eta \text{ (Efficiency)} = \frac{P_{Load}}{P_{input}} = \left(\frac{P_{input} - P_{stuff}}{(P_{input})}\right)$$

eta





$$-10V + 30I = 0$$

$$30I = 10$$

$$I = \frac{10}{30}$$

$$\boxed{I \approx 0.33A}$$

<u>Power dissipated in R_1</u>

$$P_1 = (I)^2(R_1)$$

$$P_1 = (0.33)^2(20)$$

$$P_1 = 2.178\,W$$

<u>Power dissipated in R_L</u>

$$P_L = (I)^2(R_L)$$

$$P_L = (0.33)^2(10)$$

$$P_L = 1.089\,W$$

$$P_{total} = P_1 + P_L$$

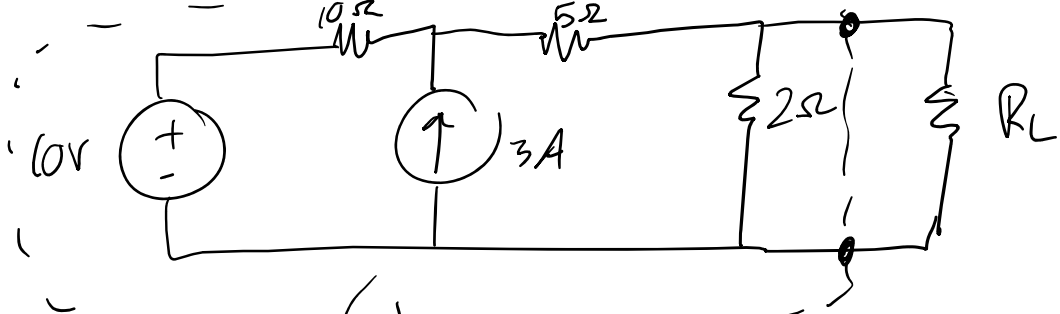
$$= 2.178 + 1.089$$

$$= 3.267\,W$$

$$\eta = \frac{P_L}{P_{Total}} = \frac{1.089\,W}{3.267\,W} = \boxed{0.33}$$

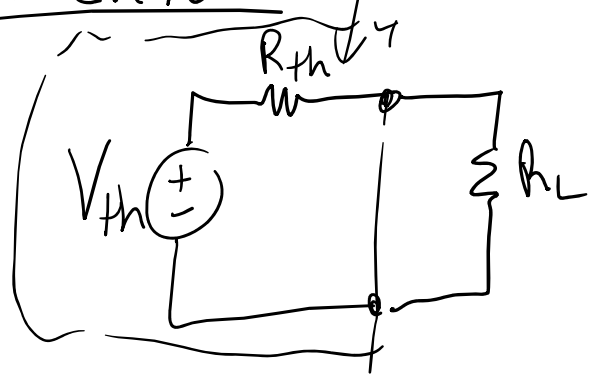
to put η in percentages multiply by 100

$$\eta = \frac{P_L}{P_{total}}(100) = \left(\frac{1.089}{3.267}\right)100 = \boxed{33\%}$$



take the crazy and turn it into something simple.

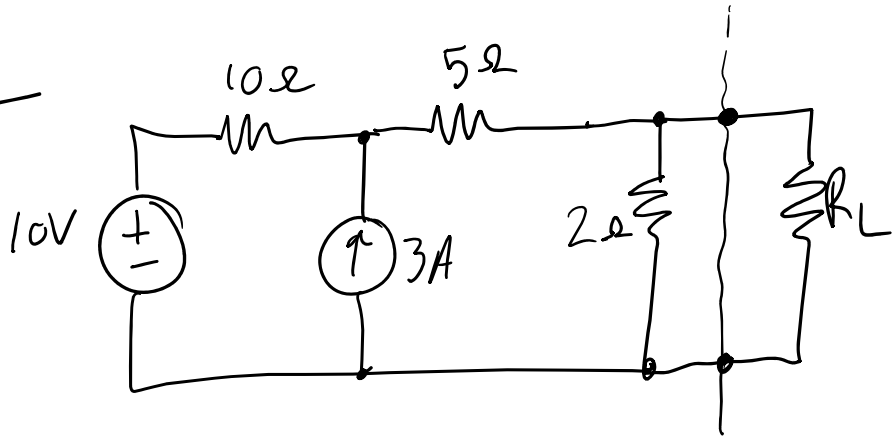
Thevenin



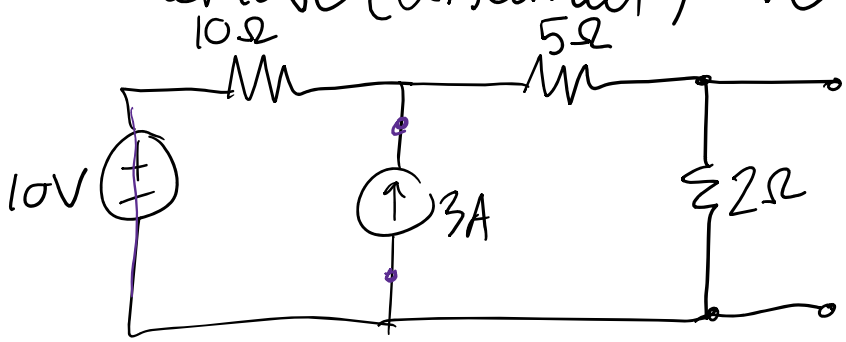
- R_{th} is the thevenin equivalent circuit
- V_{th} is the thevenin Voltage equivalent

How to find Thevenin Resistance (R_{th})

Exp

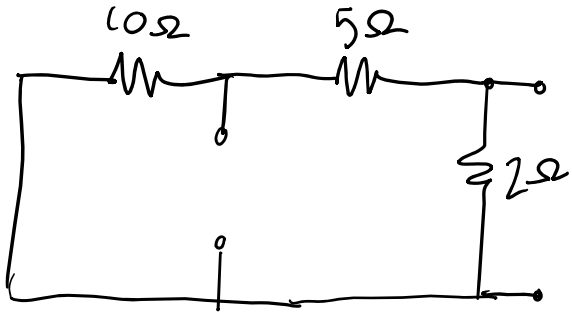


Step 1 Remove (disconnect) the Load Resistor

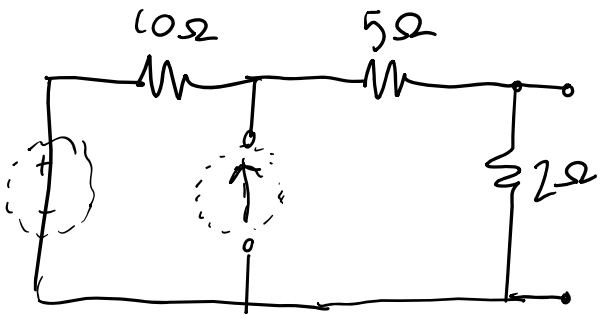


Step 2

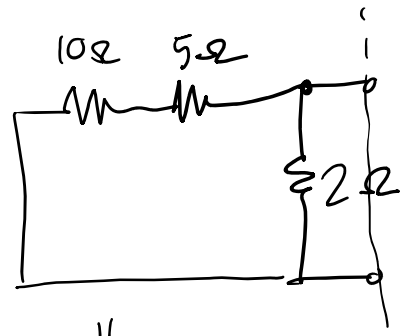
- Replace Voltage Sources with a short !
- Replace current Sources with an open



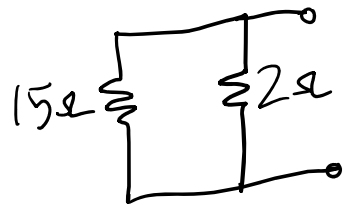
Step 3 Reduce the Resistor to a single Resistance → is R_{th}



$\Rightarrow$

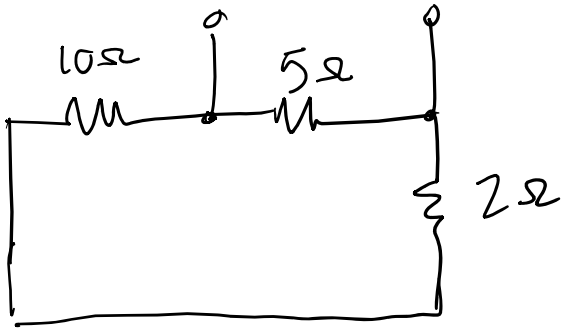


$\Downarrow$



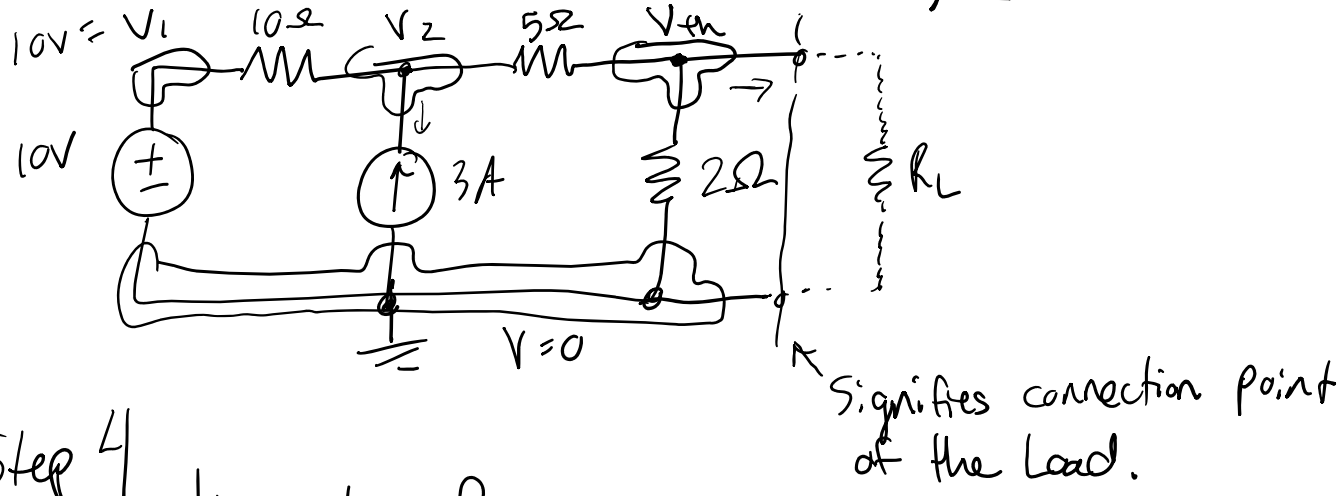
$$R_{th} = \frac{1}{\left(\frac{1}{15}\right)+\left(\frac{1}{2}\right)}$$

$$\boxed{R_{th} = 1.765\,\Omega}$$



* Keep track of where the <u>Load connection Point is!</u>

How to find Thevenin Voltage



Step 4

Keep the R_L Removed

Nodal (V_2)

$$\frac{V_2 - 10V}{10} + \frac{V_2 - V_{th}}{5\Omega} + (-3A) = 0$$

$$\frac{V_2}{10} - 1 + \frac{V_2}{5} - \frac{V_{th}}{5} = 3$$

$$\left(\frac{1}{10} + \frac{1}{5}\right)V_2 - \frac{V_{th}}{5} = 4$$

$$(0.3)V_2 - (0.2)V_{th} = 4$$

Vth Node

$$\frac{V_{th} - V_2}{5} + \frac{V_{th}}{2} = 0 \Rightarrow (-0.2)V_2 + (0.7)V_{th} = 0$$

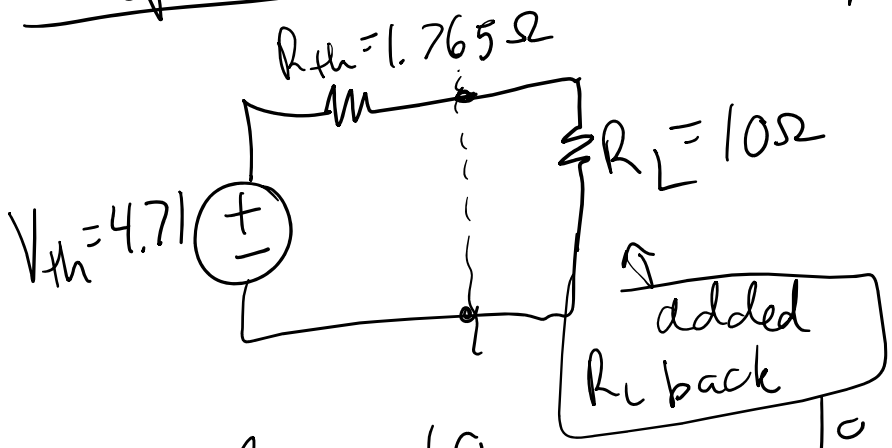
Solving for Just V_{th}

$$(0.3)V_2 - (0.2)V_{th} = 4 \quad \ldots\ldots \text{eq } 1$$

$$(-0.2)V_2 + (0.7)V_{th} = 0 \quad \ldots\ldots \text{eq } 2$$

$$\boxed{V_{th} = 4.71\,V}$$

Step 5 Redraw Circuit



$$\eta = \frac{P_L}{P_{total}} = \frac{P_L}{P_L + P_{th}} = \frac{I^2 R_L}{I^2 R_L + I^2 R_{th}}$$

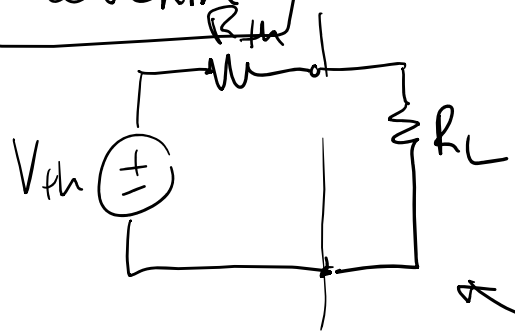
$$\eta = \frac{\cancel{I^2} R_L}{\cancel{I^2}(R_L + R_{th})}$$

$$\boxed{\eta = \left(\frac{R_L}{R_L + R_{th}}\right)}$$

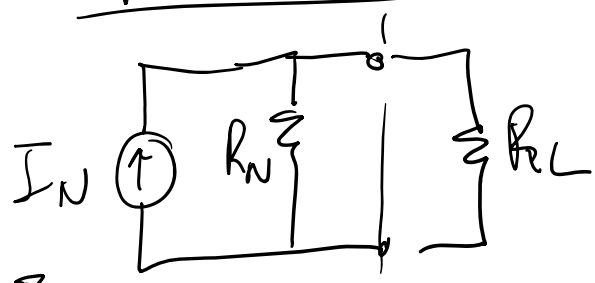
$$\eta = \frac{10}{10 + 1.765\,\Omega} = \frac{10}{11.765}$$

$$\eta = 85\%$$

* Thevenin



Norton



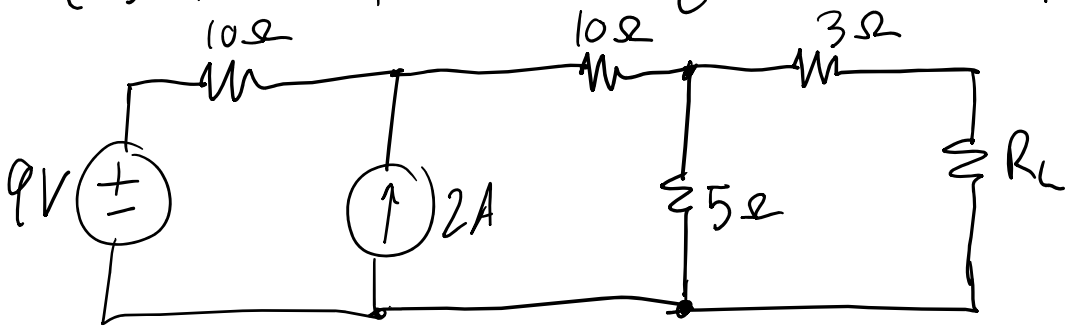
convert back and forth

$$\boxed{R_N = R_{th}}$$

$$\boxed{I_N = \frac{V_{th}}{R_N} = \frac{V_{th}}{R_{th}}}$$

Group Problem

(a) Find thevenin Equivalent of the following circuit



(b) Calculate efficiency if $R_L = 5\,\Omega$